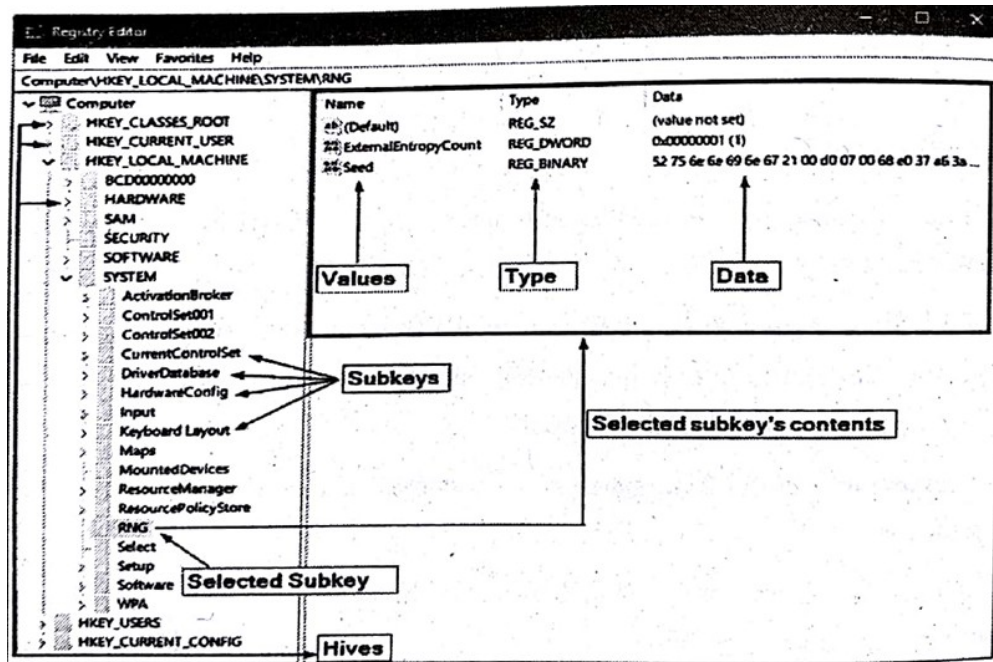


Cyber Forensics / Digital Forensics Mid 2 Solution

1. Draw and explain windows registry structure.



The **Windows Registry** is a **hierarchical database** used by the Windows operating system to store system configuration, hardware settings, user preferences, and application information. The structure is similar to a tree, consisting of hives, keys, subkeys, values, types, and data.

Components:

1. Hives

- The top level components of the registry are called Hives.
- They contain major categories of system configuration.
- Each hive stores a specific type of configuration information.
- **Examples:**
 1. HKEY_CLASSES_ROOT
 2. HKEY_CURRENT_USER
 3. HKEY_LOCAL_MACHINE
 4. HKEY_USERS
 5. HKEY_CURRENT_CONFIG

2. Keys and Subkeys

- Keys are similar to folders in a file system.
- Subkeys are keys located inside another key.
- These organize registry information in a hierarchical way.
- **Example:**

```
HKEY_LOCAL_MACHINE
├── SYSTEM
│   ├── ControlSet001
│       └── HardwareConfig
```

3. Selected Subkey

- When a subkey is selected in the left panel, its contents appear on the right side of the Registry Editor.

The diagram highlights a selected subkey (RNG) under:

- HKEY_LOCAL_MACHINE\SYSTEM\RNG

4. Values

- Values are the individual entries stored inside a key.
- They contain the actual configuration settings.
- **Examples:**
 1. (Default)
 2. EntropyCount
 3. Seed

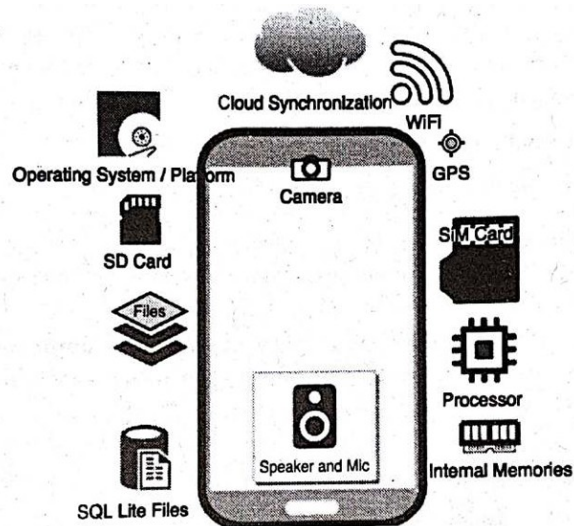
5. Type

- The Type column indicates the format of the stored data.
- **Examples:**
 1. REG_SZ – String value
 2. REG_DWORD – 32-bit numeric value
 3. REG_BINARY – Binary data

6. Data

- The Data column contains the actual value of the registry entry.
- **Example:**
 1. EntropyCount = 0x00000001 (1)

2. Discuss the basic structure of a mobile device.



A **mobile device** such as a smartphone **consists of several hardware and software components** that work together to perform communication, storage, and computing tasks.

Components:

1. **Operating System / Platform:** The operating system (OS) controls the overall functioning of the mobile device and manages hardware resources, applications, memory, and user interactions. **Examples** include Android and iOS.
2. **Processor (CPU):** Responsible for executing instructions, processing data, and controlling the operations of applications and the operating system.
3. **Internal Memory:** Stores the OS, installed applications, and user data, and typically includes RAM and flash storage.
4. **Subscriber Identity Module (SIM) Card:** Connects the device to the mobile network and stores information such as subscriber identity, network authentication information, and contacts.
5. **External Storage:** Provides additional storage space for files such as photos, videos, documents, and applications. **Examples** include SD cards and external HDDs / SSDs.
6. **Camera:** Used to capture photos and videos.
7. **Speaker and Microphone:** Handle audio input and output, where the microphone records sound and the speaker plays audio for calls, music, and notifications.
8. **Wireless Connectivity:** Mobile devices support wireless technologies such as WiFi, GPS, and cloud synchronization with services like Google Drive or iCloud.
9. **Files and Databases:** Mobile devices store information as files and databases, and many applications use SQLite databases to store data such as messages, contacts, application data, and logs.

3. What is Mobile forensic? Why it is needed.

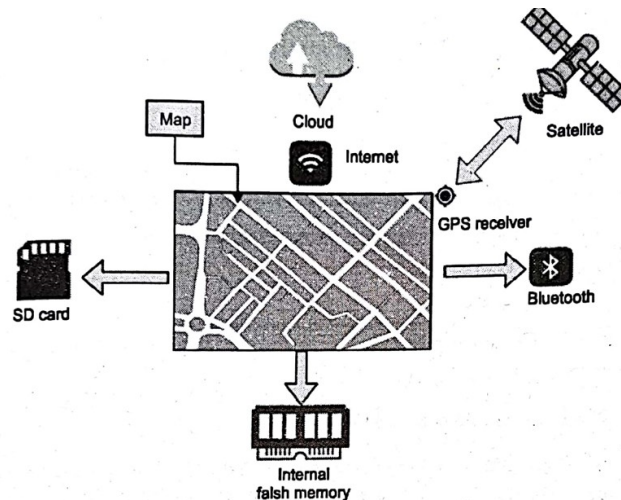
Mobile forensics is a branch of digital forensics that involves the collection, preservation, analysis, and presentation of digital evidence from mobile devices. It helps investigators recover and analyze data like call logs, messages, photos, GPS data, and application information.

Why Mobile Forensics is Needed:

1. **Crime Investigation:** Helps find evidence related to crimes such as fraud, cybercrime, and terrorism.
2. **Recovery of Deleted Data:** Allows recovery of deleted messages, files, and other information.
3. **Legal Evidence:** Provides reliable digital evidence that can be presented in court.
4. **User Activity Tracking:** Helps analyze call logs, messages, and location history.
5. **Communication Analysis:** Identifies contacts and interactions between suspects.

4. Explain GPS Forensic with diagram.

GPS forensics is a branch of digital forensics that involves the collection, preservation, and analysis of location data from GPS-enabled devices such as smartphones, navigation systems, vehicles, and wearable devices. It helps investigators determine where a device was located at a specific time and reconstruct movements related to an incident.

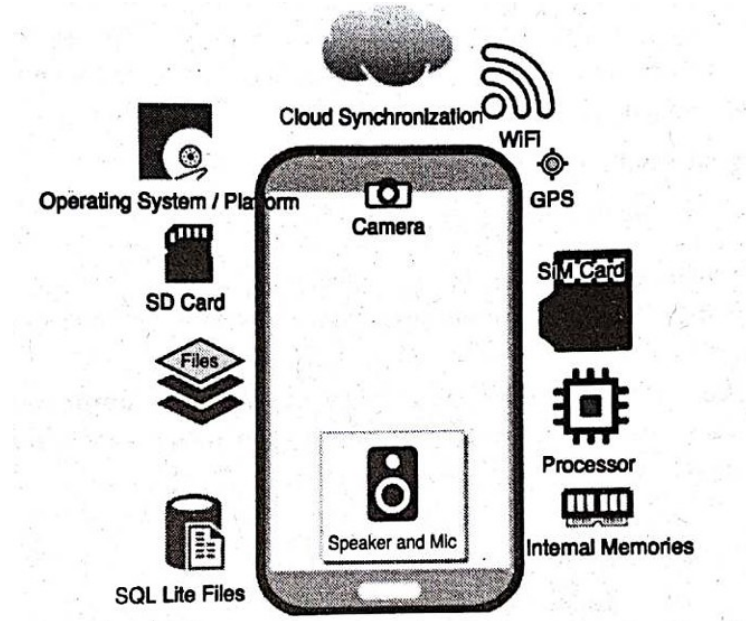


Components:

1. **Satellite:** GPS satellites orbit the Earth and transmit location signals. These signals contain time and positioning information.
2. **GPS Receiver:** Receives signals from multiple satellites. Using these signals, the device calculates its exact geographic location (latitude and longitude).
3. **Map Application:** The calculated location is displayed on map applications used for navigation and tracking.
4. **Internet / Cloud:** GPS data can be transmitted through the internet and stored in cloud services.
5. **Internal Flash Memory:** Location information is stored in the device's internal memory.
6. **SD Card:** Some devices store maps, logs, or navigation data on an SD card.
7. **Bluetooth:** GPS information may also be shared with other devices using Bluetooth.

5. Explain Mobile Forensic with diagram.

Mobile forensics is a branch of digital forensics that involves the collection, preservation, analysis, and presentation of digital evidence from mobile devices such as smartphones, tablets, SIM cards, and memory cards. It helps investigators recover and analyze data like call logs, messages, photos, GPS data, and application information.



6. What are the various hard drive analysis techniques?

1. **Data Carving:** Data carving searches raw data for file signatures to recover files, usually from unallocated space, even when file metadata is missing.
2. **Recovering Deleted Files using EnCase:** Tools like EnCase and FTK can recover deleted files. In EnCase, investigators can use the “Deleted Files” filter or File Finder to locate deleted files or specific file types.
3. **Recovering Deleted Files using FTK:** FTK Imager can recover deleted files by adding the physical drive as evidence, locating the file in the evidence tree, and exporting it to another location.
4. **Recovering Deleted Files using Fatback (Linux):** Fatback is a Linux forensic tool used to recover deleted files from disk images or storage devices.
5. **Recovering Deleted Files using Foremost (Linux):** Foremost is a recovery tool that restores files by searching for file headers and footers in forensic images.
6. **Unallocated, Free, and Slack Space Analysis:** Investigators analyze disk areas such as slack space, unallocated space, and free space to recover remnants of deleted or hidden data.

7. What are the various router investigation techniques?

1. **Establishing a Router Connection:** The investigator first establishes a direct connection to the router, preferably through the console port, to collect volatile information without alerting an attacker on the network.
2. **Recording System Time:** After connecting, the router's system time is recorded to help correlate logs and evidence later.
Example command: show clock
3. **Determining Who is Logged On:** The investigator checks currently logged-in users to identify any unauthorized access.
Example command: show users
4. **Determining the Router's Uptime:** The router's uptime shows how long it has been running since the last reboot, which can help determine when an attack may have occurred.
Example command: show version
5. **Determining Listening Sockets:** Investigators check for open ports or remote login services such as Telnet, which attackers may exploit.
6. **Reviewing the Routing Table:** The routing table is examined to understand how the router forwards packets and to detect any suspicious routing changes.
Example command: show ip route
7. **Checking Interface Configurations:** The router's network interface settings are reviewed to identify abnormal configurations.
Example command: show ip interface
8. **Viewing the ARP Cache:** The ARP cache is analyzed to view IP and MAC address mappings and detect possible spoofing attacks.
Example command: show ip arp

8. What are the various malware analysis techniques?

1. **PeStudio:** PeStudio is used for initial malware triage to identify suspicious artifacts, generate file hashes, check detections with VirusTotal, extract strings, domains, and IP addresses, and analyze entropy to detect packed or obfuscated malware.
2. **Process Hacker:** Process Hacker displays running processes on a system and helps detect processes created by malware while allowing analysts to inspect process memory for strings such as IP addresses, domains, and user agents.
3. **Process Monitor (ProcMon):** Process Monitor records real-time system activity such as file operations, process creation, and registry changes, helping analysts identify executed programs and their parent-child relationships.
4. **ProcDot:** ProcDot creates a graphical visualization of malware behavior by using Process Monitor logs to show the sequence of malicious actions.
5. **Autoruns:** Autoruns lists programs that run automatically during system startup and helps detect malware persistence mechanisms and new startup entries.
6. **Fiddler:** Fiddler is a web debugging proxy that analyzes HTTP and HTTPS traffic to identify communication between malware and command-and-control servers.
7. **x64dbg:** x64dbg is a debugging and reverse engineering tool that allows analysts to step through malware code and examine functions and API calls.
8. **Ghidra:** Ghidra is a reverse engineering tool developed by the NSA that works as a disassembler and decompiler to convert compiled malware code into human-readable form.

9. What is Windows Forensic Analysis?

Windows forensic analysis is the process of collecting, preserving, examining, and analyzing digital evidence from a Windows operating system during a forensic investigation. It involves analyzing system components such as the file system, Windows registry, event logs, user activities, and application data to detect unauthorized access, recover deleted files, and understand system behavior. The main goal is to reconstruct events and gather reliable digital evidence related to cyber incidents or criminal activities.

10. What are the Forensic Artifacts?

Forensic artifacts are pieces of digital information or traces left behind by user activities and system processes on a computer system. These artifacts are stored in locations such as the file system, Windows registry, log files, browser history, and application data. During a forensic investigation, analysts examine these artifacts to reconstruct events, track user actions, detect suspicious activity, and collect digital evidence related to an incident.

11. List and explain at least five data acquisition methods for mobile phones.

1. **Manual Acquisition:** Collects data by manually navigating through the phone interface and recording information such as contacts, messages, call logs, and photos.
2. **Logical Acquisition:** Extracts data through the device's operating system using forensic tools. It retrieves files such as contacts, SMS, call logs, and application data but usually cannot recover deleted data.
3. **Physical Acquisition:** Creates a bit-by-bit copy of the entire device memory. It allows investigators to recover deleted files and analyze hidden data.
4. **File System Acquisition:** Extracts the complete file system structure of the device, including directories, files, and system data. It provides more information than logical acquisition but less than physical acquisition.
5. **SIM Card Acquisition:** Extracts data directly from the SIM card, including contacts, SMS messages, and subscriber information stored on the card.

12. What are the 5 rules of the evidence? Explain in detail.

In **digital forensics**, evidence must follow certain rules to ensure it is acceptable in court. **The five rules of evidence are:**

1. **Admissible:** Evidence must be legally obtained and allowed in court according to the law. If the evidence is collected without proper authorization or violates legal procedures, it may be rejected by the court. Therefore, investigators must follow proper forensic procedures and legal guidelines when collecting digital evidence.
2. **Authentic:** Evidence must be proven to be genuine and related to the incident under investigation. The investigator must demonstrate that the evidence has not been altered and that it originated from the specific device or system being examined.
3. **Complete:** Evidence must present the entire picture of the incident. Investigators should collect all relevant data so that the court can clearly understand what actually happened.
4. **Reliable:** Evidence must be collected and analyzed using reliable forensic tools and methods. The procedures used should produce consistent and accurate results so that the evidence can be trusted.
5. **Believable:** Evidence must be clear and understandable to the court or jury. The investigator should present the findings in a way that non-technical people can easily understand.

13. What is a file system?

A **file system** is a method used by an operating system to organize, store, manage, and retrieve data on storage devices such as hard drives, SSDs, and USB drives. It defines how data is structured into files and directories and how these files are stored and accessed on the disk. The file system also keeps track of information such as file names, locations, sizes, permissions, and timestamps. **Common examples** of file systems include FAT, NTFS, exFAT, and EXT.

14. Write a brief note on Reviewing Pertinent Logs in UNIX.

In UNIX systems, **log files provide important information during incident response and forensic investigations**. These logs record various system activities such as user logins, system startups, shutdowns, and network service events. During an investigation, analysts review these logs to identify suspicious activities, system errors, or unauthorized access attempts. **Most UNIX log files are stored in the /var/log directory**, which contains records of system operations and security events. In some UNIX variants, logs may also be found in directories such as /usr/adm or /var/adm. Occasionally, certain log files may be located in other directories like /etc.

15. Write a note on Windows file system?

The **Windows file system** is the structure used by the Windows operating system to store, organize, and manage data on storage devices such as hard drives and SSDs. It controls how files and directories are created, accessed, stored, and retrieved. **Windows mainly uses the NTFS (New Technology File System)**, although older systems also support FAT and FAT32. NTFS provides advanced features such as file permissions, encryption, compression, and journaling, which improve security and reliability. The Windows file system **organizes data into files and folders (directories)**. It also maintains metadata such as file name, size, location, creation time, modification time, and access permissions.

16. Write a note on UNIX file system?

The **UNIX file system** is a hierarchical structure used by UNIX operating systems to organize and manage files and directories on storage devices. It **stores data in a tree-like structure that begins with the root directory (/)**, from which all other directories and files branch out. In the UNIX file system, **everything is treated as a file**, including devices and processes. Important directories include /bin (system binaries), /etc (configuration files), /home (user directories), /var (log files), and /tmp (temporary files). The UNIX file system **manages data using inodes**, which store metadata such as file permissions, owner information, size, and timestamps. This structure helps the operating system efficiently locate and manage files.

17. Write a brief note on Windows Recycle Bin Forensics.

The **Windows Recycle Bin** stores files that have been deleted by users but are not permanently removed from the system. Instead of deleting a file immediately, Windows moves it to the Recycle Bin, allowing it to be restored later. This feature was introduced in Windows 95 and is the default behavior when a user deletes a file using the Delete key or the context menu. However, files can bypass the Recycle Bin if the user presses Shift + Delete or changes system settings.

In modern Windows systems, deleted files are stored in the \$Recycle.Bin folder located in the root directory of the drive (e.g., C:\$Recycle.Bin). Inside this folder are subfolders named with the user's Security Identifier (SID), which represent individual user accounts. Each user's folder contains the files they have deleted.

For each deleted file, two types of files are stored:

- **\$R files:** Contain the actual deleted data that can often be recovered.
- **\$I files:** Contain metadata about the deleted file, such as the original file path, file size, and deletion time.

During forensic analysis, investigators examine the \$Recycle.Bin directory and the SID folders to identify which user deleted the file. Tools such as \$I Parser can be used to extract metadata from \$I files and generate reports (e.g., CSV files) containing information about deleted files.

18. State the difference between investigation and forensics.

Investigation	Forensics
Investigation is the process of searching for facts and information related to a crime or incident.	Forensics is the scientific analysis of evidence collected during an investigation.
It focuses on identifying suspects, motives, and events .	It focuses on examining and analyzing evidence using scientific methods.
It may involve interviews, surveillance, and gathering information .	It involves technical examination of digital or physical evidence .
The goal is to understand what happened and who is responsible .	The goal is to produce reliable evidence that can be presented in court .
Investigation is a broader process that includes many activities.	Forensics is a specialized part of the investigation process .

19. Explain the concept of “Trust Relationships”.

A **trust relationship** refers to a security relationship between two systems, domains, or networks in which one system trusts another to authenticate users and allow access to its resources. In such a relationship, a user who is authenticated in one domain can access resources in another trusted domain without needing to log in again.

Trust relationships are commonly used in Windows domain environments to allow resource sharing between different domains or networks. They help organizations manage authentication and authorization across multiple systems, making it easier for users to access shared files, applications, and services.

However, trust relationships can also create security risks. If one trusted system is compromised, an attacker may gain unauthorized access to other systems that trust it. Therefore, proper monitoring and security controls are necessary to manage trust relationships safely.

20. In android terms, what is rooting?

Rooting in Android is the process of gaining root (administrator or superuser) access to the Android operating system. It allows users to obtain full control over the device and modify system files and settings that are normally restricted by the manufacturer.

By rooting an Android device, users can install custom ROMs, remove pre-installed applications, modify system settings, and run specialized applications that require administrative privileges. However, rooting may void the device warranty, reduce system security, and potentially damage the device if done incorrectly.

21. What is JTAG?

JTAG (Joint Test Action Group) is a hardware interface and standard used to access and control the internal components of electronic devices such as mobile phones, routers, and embedded systems. It was originally designed for testing and debugging circuit boards during manufacturing.

In digital forensics, JTAG is used as a data acquisition method to extract data directly from a device's memory. This allows investigators to obtain a complete copy of the memory contents, including deleted or hidden data, without relying on the device's operating system.

22. What are the two most popular applications of GPS?

1. **GPS Tracking:** Used to track the location and movement of a device, vehicle, or person in real time.
2. **GPS Navigation:** Used to provide directions and route guidance from one location to another using satellite signals.

23. List four types of evidence that may found within GPS device.

1. **Location Data:** Latitude and longitude coordinates showing the device's position.
2. **Route History:** Records of routes or paths previously traveled by the device.
3. **Waypoints / Saved Locations:** User-saved destinations or frequently visited places.
4. **Time and Date Information:** Timestamps associated with location data that show when the device was at a specific place.

24. Why SIM card forensics is important?

SIM card forensics is important because a SIM card stores valuable information related to a mobile user's communication and identity. It contains data such as subscriber identity (IMSI), contacts, SMS messages, call records, and network information. During a forensic investigation, analyzing the SIM card helps investigators identify the user of the device, recover communication records, trace connections between suspects, and gather evidence related to criminal activities. Even if the phone is damaged or unavailable, the SIM card may still provide useful digital evidence.

25. What are PIN and PUK numbers?

PIN (Personal Identification Number) is a security code used to protect access to a SIM card in a mobile phone. When the phone is turned on, the user must enter the PIN to unlock the SIM and use network services such as calls and messages.

PUK (Personal Unblocking Key) is a special code used to unlock the SIM card if the PIN is entered incorrectly multiple times and the SIM becomes blocked. The PUK is usually provided by the mobile network operator to restore access to the SIM.